

SAFE WORK METHOD STATEMENT (SWMS)

In order of priority:

- **ELIMINATION** – Remove hazard completely.
- **Substitution** – Replace object /process with less hazardous one.
- **Isolation** – Isolate the hazard (i.e. separate the hazard from the people).
- **Engineering** – Modify a process (i.e. redesign a workstation).
- **Administration** – Use procedures /systems to minimise the risk (i.e. Rotation of employees, implementation of a safe work method, training, etc).
- 6. **Personal Protective Equipment (PPE)** – use to try to protect the person.

RISK SCORE CALCULATOR			
High	High	Extreme	Extreme
Mod.	High	Very High	Extreme
Mod.	High	Very High	Very High
Mod.	Mod.	High	Very High
Low	Low	Mod.	High
MINOR	MODERATE	FATAL	CATASTROPHIC

Residual Risk	Action Required
Low	Ok for now. Continue to monitor.
MODERATE	Determine if remedial action/treatment is required.
HIGH	Immediate decision required by site leadership. Identify and commence remedial action with appropriate priority.
VERY HIGH	ACT NOW. Requires immediate attention by site leadership. Address the risk without delay. Requires urgent attention and clear, timely action plans to reduce the risk before proceeding.
EXTREME	ACT NOW. Requires immediate attention by site leadership. Urgently address the risk. Requires immediate resolution with focused attention until resolved.

Identify task high risk workplace hazards - *Non routine work is work not documented in a Safe Operating Procedure (SOP)* or Work Instruction

Note: If machine isolation is required, workers must be competent in the relevant isolation procedure.

EXTREME & VERY HIGH RISK HAZARDS	Risk	MAIN RISKS / PERMITS REQUIRED
1. Entering a Confined Space.	E	- Heat exhaustion, engulfment, limited access/egress. - Asphyxiation – low O₂ or contaminants. (Confined Space Entry Permit required) - Explosive atmosphere, combustible dusts (Confined Space Entry/Hot Work Permits required)
2. Work in an area which may have a contaminated or flammable atmosphere.	E	- Fire / explosion from ignition source. (Hot Work Permit required) - Asphyxiation or poisoning from contaminated atmosphere. (Confined Space Permit)
3. Work with potential for a fall from 2m or work on a roof	E	Falling worker/object ≥2m requires SWMS.
4. Hot work required. (Oxy cutting, welding, soldering, grinding, etc).	E	- Fire or explosion from ignition source. (Hot Work Permit required) - Injury caused by hot surface, grinding wheel ejected pieces, welders "flash", etc. Fire inside Expanded Polystyrene (EPS) wall panels (Hot Work Permit required).
5. Hand dig >200mm OR ground penetration >200mm OR mechanical earthworks OR enter a trench >1.5 M deep (excavating is Construction work)	E	- Contact utilities – electricity, water, gas, communications. (Excavation Permit required) - Trench Collapse. (Excavation Permit required)
6. Non routine electrical, high voltage or 'live' work or work near electrical installations.	E	- High voltage is >33kV. 'Live' work is when electrical equipment is worked on energised – (Management Approval Required) - Electrical cabinets or MCC identified as Red
7. Non routine work involving mechanical lifting, lowering, pushing or pulling.	E	- High mechanical force damage / injury. - High risk work licence for use of cranes, mobile plant, scaffold, dogging, etc.
8. Using high risk tools with excess noise, kickback, cutting edge, high speed/pressure, ejected material, weight, awkward posture, explosive force.	E	- Injury from use of high risk powered or hand tools. - Electrocution, burn or electric shock. - Manual task injury from weight, forces, overexertion, repetition or poor work heights.
9. Demolish / alter load bearing / integral part of a structure 6m high OR use load shifting machinery on a suspended floor OR use explosives. Includes modify handrail / floors.	E	Being struck or crushed by parts of a structure or exposure to asbestos, lead, hazardous chemicals etc. (Demolition License and/or asbestos removal licence and Safe Work Plan required)
10. Work inside fences / barricades / enclosures where moving / rotating machinery is operating or hazards may exist.	V	- Serious bodily injury caused by contacting moving/unguarded plant or materials. - Machinery starting unexpectedly or electrocution, burn or electric shock.
11. Potential or stored energy which cannot be isolated (electrical, hydraulic, pneumatic, kinetic).	V	- Electrocution or entrapment. - Machinery entanglement / injury.
12. Work on or near a pressurized or enclosed circuit (water, pneumatic, hydraulic, hot oil, etc).	V	- Fracturing the mains or piping from contact – spills / leaks / emissions (environment). - Contacting temperature extremes.
13. Working in artificial temperature extremes or weather extremes. Hot, cold, wet, icy, lightning, high winds.	V	- Electrocution from using power tools / leads in wet conditions or lightning. - Heat stroke / Heat Exhaustion / Fatigue / Frost bite / Hypothermia / Influenza. - Slip / Trip / Fall Hazards.
14. Hazardous manual tasks or excessive noise or excessive vibration.	V	Personal injury/illness or hearing loss.
15. Using a hazardous chemical – chemicals, heavy metals, asbestos, Lead.	V	- Exposure to hazardous chemicals. - Spills / leaks (environmental).

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16. Work on or adjacent to road, rail line or water body.	V	• Being hit by passing traffic or drowning.
17. Work on or near a chemical, glue, fuel, oil or refrigerant line / system.	V	• Contact chemicals (contamination, injury, etc). • Fracturing the line from contact – fire or explosion from ignition source – spill / leak.
18. Changes to software control system.	V	• Change to way plant / equipment operates & workers not retrained.
19. Other (Specify):		

SWMS Title: Confined Space Entry – Welding Repair of Cracks in Dryer Drum

Prepared By: Chris Foster

Date: 23-Jun-2026

Business Name: Tuan Biomass Albioma

Site: Tuan Biomass

Operational From: 23/06/2026 7:00 am **To:** 24/06/2026 2:30pm

Description of Task: Isolation and gas testing of dryer drum, confined space entry, and gas metal arc/MIG weld repair of identified cracks on the internal drum shell. Includes pre-entry inspection, atmospheric monitoring, hot work permit controls, fire watch, and post-work inspection prior to return to service.

Work/Project Location: Dryer Drum, Dehydration Line.

Reason for analysis: ☐ New task/plant ☒ High Risk Task ☐ Change ☐ Regular review ☐ Post-Incident ☐ Contractor (to complete) Task ☐ Non-Routine Task ☒ Maintenance/Repair ☐ Other (specify):

SWMS Team Names: 1. Chris Foster

2.

3.

4.

5.

6.

Permits Required: (circle)

Confined Space ☒ Risk of Falls ☐
Excavation ☐ Hot Work ☒

PPE Requirements: (circle)



Potential Environment Hazards (circle)

Fire/Emergency Response Equipment

Special tools or Equipment Required (e.g. gas detection, ventilation fans, lighting, scaffolding etc.) List: 4-gas detector, fire extinguisher, two-way radios

Noise pollution ☐ Soil erosion ☐
Air pollution dust/smoke ☐ Hazards to flora & fauna ☐
Spills to ground ☐ Spills to water ☐

1. First Aid Kit: Smoko Room
2. Communication method/details
Verbal, 2way radio,
3. Fire hose
4. Other: _____

Hazards Identified

Electricity ☐	Hazardous Chemicals ☐	Pressure (air, water, gas) ☐	Biological ☐	Hot Work ☐	Person/Object Falling ☐
Machinery & Equipment ☐	Lighting ☐	Ignition Sources ☐	Psychological ☐	Confined Space ☐	Traffic/Mobile Plant ☐
Hazardous Manual Tasks ☐	Dust and or fume ☐	Extreme Temperatures ☐	Radiation ☐	Noise ☐	Excavation/ Penetration ☐
Other workers/people					Stored Energy ☐

Other Comments

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Key Steps in Logical Order		Hazards/Risks to Workers or Public	Risk Score	Control Measures	Residual Risk Score
1	Permit, isolation & preparation	Planning – entering and welding inside a confined space (dryer drum) without authorisation, isolation or atmosphere testing. Engulfment, asphyxiation, explosion.	Med	Complete SWMS toolbox talk with all crew. Confined Space Entry Permit, Hot Work Permit, isolation permit authorised by Tuan Biomass. isolate and lock out/tag out dryer drum drive, burner, verify isolation with try-test. Establish exclusion zone and signage at entry point.	Low
2	Atmospheric testing & ventilation	Atmospheric hazards inside drum – low oxygen, combustible dust, residual gases. Asphyxiation, fire/explosion.	Med	4-gas detector to test atmosphere before entry and continuously monitor throughout entry, including during welding. ventilate drum with ID fan prior to and during work. Do not enter if readings are outside safe limits; re-test after any break in work. Record results on Confined Space Entry Permit.	Low
3	Confined space entry & standby	Entry/egress restricted, fall or slip on internal surfaces, communication failure between entrant and standby person. Delayed rescue.	Med	Appoint dedicated standby person (Confined Space Attendant) at entry point at all times work is in progress; standby person not to enter drum. Establish agreed communication method visual or radio check every 15 minutes.	Low
4	Hot work – weld repair	Hot work – MIG/arc welding to repair cracks. Fire from ignition source, burns, fumes, electric shock, arc flash.	Med	Hot Work Permit in place; remove or shield combustible material within drum and surrounding area. Position fire hose and extinguisher and at entry point; Tuan Biomass nominate dedicated fire watch for duration of welding and 30 and 60 minutes after finished. Entrant to wear all appropriate PPE. OZ tac Inspect welding leads and earth return for damage before use.	Low
5	Ongoing entry – fatigue/heat management	Manual handling and fatigue/heat stress working in a confined, hot drum environment for extended periods.	Med	Rotate work at regular intervals. Monitor entrants for signs of heat stress, fatigue or fume exposure; stop work immediately if symptoms develop. Use mechanical aids/trolleys to move welding equipment and consumables rather than manual carrying where practicable.	Low

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6	Weld inspection & quality check	Weld repair quality – inadequate repair leading to future structural failure of drum shell.	Med	Welder to be appropriately ticketed/qualified for the weld procedure used. Grind back cracks to sound metal and visually inspect before welding. Carry out post-weld visual inspection to confirm repair integrity. Remove all slag, spatter and ignition sources from drum interior before closing up.	Low
8	Job completion & return to service	Return to service – drum re-started with tools, materials or personnel still inside, or isolations not removed correctly.	Med	Tuan biomass standby person Conduct final headcount and visual inspection of drum interior to confirm it is clear of personnel, tools and debris before closing access. Remove confined space entry signage/barricades. Tuan Biomass isolation officer De-isolate. Test-run dryer drum and confirm normal operation before returning to normal operation.	Low

Attach additional pages if required

All relevant workers to be consulted during this process and made aware of the control measures and all required permits identified.

AUTHORISATION			
Approved by	David Knight	Position	Plant manager
Signed		Date	

Name, Sign Off & Employer of Persons Performing Work		Start Time:	am/pm	Finish Time:	am/pm
Name	Sign Off	Employer			
1					
2					
3					
4					
5					

JOB COMPLETION					
1. Has equipment been modified/changed to affect how it operates?		☐ N (go to 3) ☐ Y (sign off by competent operator required)			
Has a competent operator approved it is safe to use? ☐ Y ☐ N	Competent Operator Name:		Competent Operator Signature:		
2. Work completed & area assessed safe before return to normal service? ☐ Y ☐ N To be completed by the person conducting the task		Name:			
		Signature:			
3. Albioma Representative advised of & accepts changes: ☐ Y ☐ N		Name:		Signature:	

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